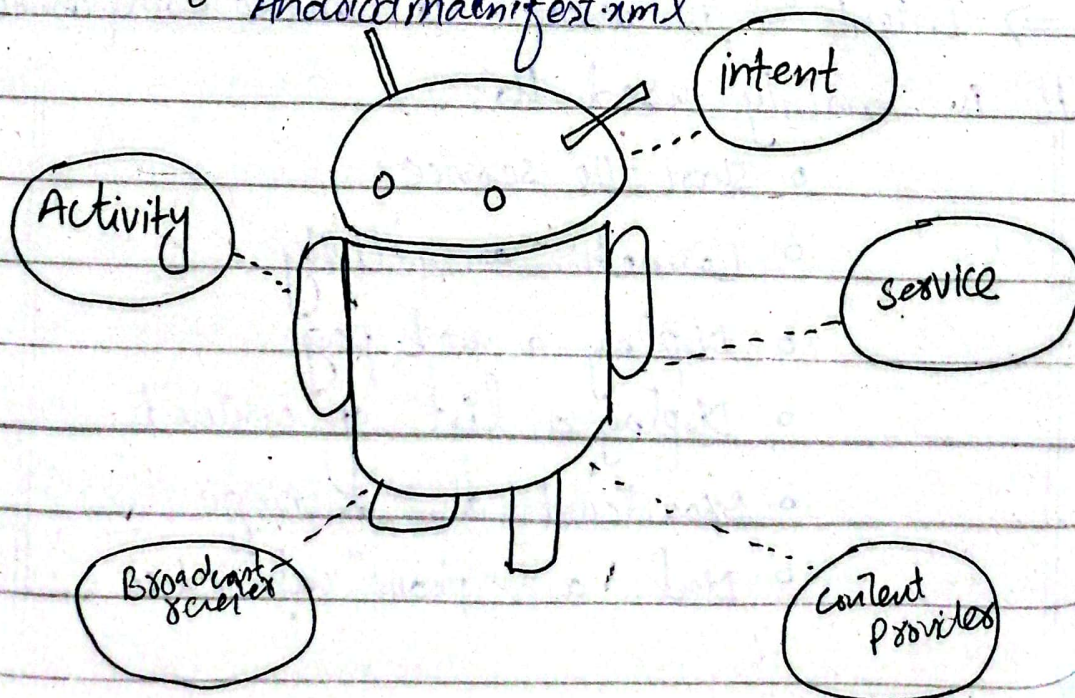


Core Building Blocks of Android

⇒ An android component is simply a piece of code that has a well defined life cycle e.g. Activity, receiver, service etc.

⇒ The core building blocks or fundamental components of android are:

- Activities
- Views
- Intents
- Services
- Content providers
- Fragments
- AndroidManifest.xml



(1) Activity: - it is Application presentation layer.
 $\Rightarrow$ An activity is a class that represents a single screen. It uses views to form graphical user interfaces that display info & respond to user actions.

$\Rightarrow$ It is like a frame in AWT.

$\Rightarrow$ simply screen of mobile.

$\Rightarrow$ It has two parts:

- o Content part (.xml) - it is similar to a window in the windows OS.
- o Backend (.java)

$\Rightarrow$ Activity = .xml + .java

(2) View: - it is a small rectangular box that responds to user input.

$\Rightarrow$ A View is the UI element such as button, label, text field etc. Anything that you see is a view.

(3) Intent: - A simple message passing framework.

$\Rightarrow$ Intent is used to invoke component.

$\Rightarrow$ It is mainly used to:

- o Start the service
- o Launch an activity
- o Display a web page
- o Display a list of contacts
- o Broadcast a message
- o Dial a phone call etc.

- Services can be used for IPC b/w apps.
 $\uparrow$ it is designed to do some work without a user interface.

Service: - invisible workers of application.

$\Rightarrow$ Service is a background process that can run for a long time.

These are two types of services: e.g. Play music, download file etc.

(1) Local

(2) Remote

Local: - is accessed from within the application.

Remote: - where remote service is accessed remotely from other applications running on the same device.

Content providers: - shareable database.
 that connects data in process with code running in other process.

Content providers are used to

share data between the applications.

Fragment: - Preferred way of sharing data across application boundaries. Such requests are handled by the methods of the ContentRes.

$\Rightarrow$ Fragments are like parts of activity.

$\Rightarrow$ An activity can display one or more fragments on the screen at the same time.

Android manifest.xml:

It contains information about activities, content, providers, permissions etc.

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Android manifest.xml: -

It contains information about activities, content, providers, permissions etc.

⇒ It is like the web.xml file in java EE.

Android Virtual Device:-

⇒ It is used to test the android application without the need for mobile or tablet etc.

⇒ It can be created in different configurations to emulate different types of real devices.

Broadcast Receivers → is a dormant comp that listens to system

⇒ Intent broadcast consumer. By creating & registering a Broadcast Receiver, an application can listen for broadcast intents that match specific filter criteria.

Notifications

⇒ A user notification framework. Notifications allow signal users without stealing focus or interrupting their current activity. It is a message that displays outside app UI to provide the user with reminders, timely info.

⇒ By decoupling the dependencies between application component, it is possible to share and interchange individual pieces such as services etc with other applications.

Q. What is android SDK?

⇒ SDK stands for software development kit.

⇒ SDK is a set of development tools which includes debugger, libraries, documentation, sample codes, tutorials, emulator (AVD) etc.

⇒ Whenever google releases new android version, it also releases the new version of SDK.

⇒ Android SDK contains three components:-

- SDK platforms
- SDK Tools
- SDK update sites

Q. What is Gradle?

⇒ Gradle is an open source build system which is used to automate building, testing, deployment etc for our application.

⇒ Gradle is also used to generate the .apk file from .java and .xml file.

⇒ Gradle (build.gradle) contains the information about min sdk version, target sdk, android app versions etc.

Q. What is android emulator?

⇒ Emulator is a virtual device which acts same as physical mobile device.

⇒ Emulator is used to test, debug and execute our android application.

Emulator is heavy device so our system configuration should be high in our emulator.

⇒ Used as target device.

⇒ It provides almost all the functionality of a real device.

⇒ We can get incoming phone calls and text messages.

⇒ It also gives the location of the device and simulates

different network speeds.

⇒ Testing android applications on emulator are sometimes faster and easier than doing on a real device.

Q. What are requirements & recommendations for emulator?

⇒ The android emulator takes some additional requirements beyond the basic system requirement for android studio.

⇒ These requirements are:

- SDK Tools 26.1.1 or higher
 - 64-bit processor
 - HAXM 6.2.1 or later
- Windows: CPU with VM

Q. What is Dalvik virtual machine?

⇒ Dalvik virtual machine is a register-based virtual machine that is part of android OS.

- The DVM executes files in the Dalvik executable (.dex) format & relies on the Linux kernel for scheduling & low-level memory management.

⇒ It was designed and written by Dan Bernstein with contributions of other Google engineers.

⇒ The Dalvik virtual machine was named after Bernstein after the fishing village "Dalvik" in Eygafjörður, Iceland where some of his ancestors to live.

⇒ it is an integral part of android software.

Features:

⇒ Optimizing the virtual machine for:

- memory
- Battery life
- Performance

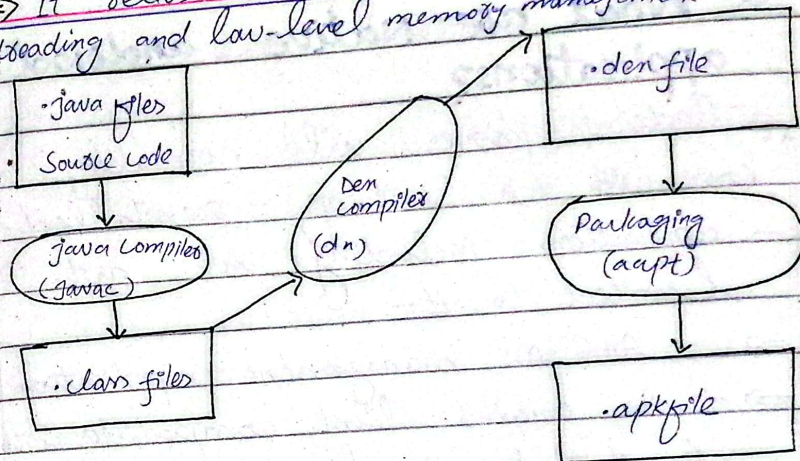
⇒ it converts class files into .dex file through Dalvik Compiler that runs on Dalvik vm.

⇒ it supports the only android OS.

⇒ In DVM executable is APK.

⇒ Optimized to ensure that a device can run multiple instances.

⇒ It relies on the Linux kernel for scheduling and low-level memory management.



Q. What are the elements of the AndroidManifest.xml file?

⇒ The elements are:

- <manifest>
- <application>
- <activity>
- <intent-filter>
- <action>
- <category>